Onail Nissan
Housing Affordability Analysis Capstone project

Power BI Modeling and Data Documentation

Contents

Data Cleaning, Merging, and Modeling – Other Queries in Power Query **2**
Household_Income_Data (2015-2024) 2
Household Income Data Estimate - 2025 4
Median_HouseHold_Income 4
Median_Home_Values 5
Linear Regression Model – Measures in Power BI **8**
Median Home Value Linear Regression 8
MedianHouseholdIncomeLinearRegression 10
Measures **12**
Affordability Ratio 12
Affordability Ratio – Regression Values 12
Home Growth % Since 2015 13
Home Growth % Overview 14
Income Growth % Since 2015 14
Down Payment (5%) 15
Down Payment (20%) 16
Data Model and Table Relationships **17**
Key Column 17
Relationship Between Household Income and Median Home Values 18
Relationship Between Year Table and Median Home Values 18

Back to Table Of Contents

Data Cleaning, Merging, and Modeling – Other Queries in Power Query

This document will explain how the data processing tasks for this project were completed using Power BI. More specifically, it will cover data cleaning, data merging, and data modeling. These tasks were completed using tools such as Power Query within Power BI instead of Python. This document will outline the steps and logic used to prepare and analyze the data in this housing affordability analysis project.

Household_Income_Data (2015-2024)

This query is used to get all of the datasets from the **Median_Household_Income** folder in the **Data** folder. It combines the data into a single table and then will apply multiple transformation steps in order to clean and structure the dataset for analysis.

Applied Steps

#1 - Lowercased Text – This step converts all of the text in the Label(Grouping) column to lowercase in order to extract the year value in **Applied Step - #2 - Extracting Year.**

- Select the **Label(Grouping)** column
- Go to the **Transform** tab
- Click **Format** and choose Lowercase.

Formula: = Table.TransformColumns(#"Expanded Transform File",{{"Label (Grouping)", Text.Lower, type text}})

#2 - Extracting Year – This step extracts the year value from the text in the **Label(Grouping)** column by using the extract the value between the text **(in** and **inflation.**

- Select the **Label(Grouping)** column
- Go to the **Transform** tab
- Click Extract and choose **Text Between Delimiters**
- Enter the following
    - **Start delimiter**: (in
    - **End delimiter**: inflation
- Click OK

Formula: = Table.TransformColumns(#"#1 - Lowercased Text", {{"Label (Grouping)", each Text.BetweenDelimiters(_, "(in", "inflation"), type text}})

Back to Table Of Contents

#3 - Renaming Two Columns – This step renames two columns **Label (Grouping)** and **ZCTA5 85382!!Estimate** for a meaningful column name.

- Right click the column one by one
- Select **Rename**
- Rename each column
    - Label (Grouping) **-> Year**
    - ZCTA5 85382!!Estimate -> **Median Household Income**

Formula: = Table.RenameColumns(#"#2 - Extracting Year",{{"ZCTA5 85382!!Estimate", "Median Household Income"}, {"Label (Grouping)", "Year"}})

#4 - Added Custom Column to Represent Zip code – This step adds a new column called **Zip Code** with the value of 85382 to all rows.

- Go to the **Add Column tab**
- Click **Custom Column**
- Name the column **Zip Code**
- Enter the value **85382** in the Custom column formula
- Click **OK**

Formula: = Table.AddColumn(#"#3 - Renaming Two Columns", "Zip Code", each 85382)

#5 - Removing Source.Name Column – This step removes the Source.Name Column

- Right click the **Source.Name Column**
- Choose **remove**

Formula: = Table.RemoveColumns(#"#4 - Added Custom Column to Represent Zip code",{"Source.Name"})

#6 - Reordering Columns – This step orders the column into Year, Zip Code and Median Household Income

- Click on each column and move the columns into the following order
- Year – Zip Code – Median Household Income

Formula: = Table.ReorderColumns(#"#5 - Removing Source.Name Column",{"Year", "Zip Code", "Median Household Income"})

#7 - Changing Zip Code into Whole Number Data Type – This step is used to change the **Zip Code** column into a whole number data type.

- In the column header for **Zip code –** click on the icon to the left of the title

Back to Table Of Contents

- Choose **whole number**

Formula: = Table.TransformColumnTypes(#"# 6 - Reordering Columns",{{"Year", Int64.Type}, {"Zip Code", Int64.Type}})

Household Income Data Estimate - 2025

This query creates a new table that has an entry for the year 2025. Since data.census.gov does not have our dataset for the year 2025, this table will be an estimate. The estimated value is made by taking the difference between the 2024 and 2024 values and then combing that value with the 2024 value. Using this method, the estimated 2025 median household income was calculated by 89,663.

- Go to the **Home Tab**
- Click **Enter Data**
- Add the following fields
- Column: **Year –** Value: **2025**
- Column: **Zip Code** – Value: **85382**
- Column: **Median Household Income** – Value: **89663**
- Name the table **Household_Income_Data Estimate (2025)** and choose **OK**

Median_HouseHold_Income

This step combined both the Household_Income_Data (2015-2024) and Household_Income_Data Estimate (2025) into one single table containing Median Household Income from 2015 – 2025.

- Go to the **Home** tab
- Select **Append Queries**
- Select **Append Queries as New**
- Choose the two tables
  - Household_Income_Data (2015-2024)
  - Household_Income_Data Estimate (2025)
- In the column header for **Median Household Income –** click on the icon to the left of the title
  - Choose **whole number**

Back to Table Of Contents

Median_Home_Values

This step removes columns that are not needed for this dataset.

#1 - Removed Columns – This step removes the multiple columns

- Hold **Ctrl** and select the following columns:
    - RegionID
    - SizeRank
    - RegionType
    - StateName
    - State
    - City
    - Metro
    - CountyName
- Right click any of the selected columns
- Choose **Remove Columns**

Formula: = Table.RemoveColumns(#"Promoted Headers",{"RegionID", "SizeRank", "RegionType", "StateName", "State", "City", "Metro", "CountyName"})

#2 - Unpivoted Columns – This step convers the dataset from the current wide format into a long format by turning the date columns into rows.

- Select the **RegionName** column
- Go to the **Transform** tab
- Click **Unpivot Other Columns**

Formula: = Table.UnpivotOtherColumns(#"#1 - Removed Columns", {"RegionName"}, "Attribute", "Value")

#3 - Filtered Rows – This step is to filter the **Zip Code** Column to the zip code focus for this project which would be **85382**.

- Select the **RegionName** column
- Click the filter dropdown
- Click on **Text Filters**
- Set to **Equals**
- Put the zip code value: **85382**
- Click **OK**

Back to Table Of Contents

Formula: = Table.SelectRows(#"#2 - Unpivoted Columns", each [RegionName] = "85382")

#4 - Renamed Columns – This step renames the **RegionName** to **Zip Code** for a meaningful column name.

- Right-click the **RegionName** column
- Select **Rename**
- Change the name to **Zip Code**

Formula: = Table.RenameColumns(#"#3 - Filtered Rows",{{"RegionName", "Zip Code"}})

#5 - Changing Column Type to Date – This renames the **Attribute** column into a data date type

- Select the **Attribute** column
- Go to the **Transform** tab
- Click **Data Type**
- Choose **Date**

Formula: = Table.TransformColumnTypes(#"#4 - Renamed Columns",{{"Attribute", type date}})

#6 - Inserting Year Column – This step creates a new column called **Year** by extracting the year value from the columns **Attribute** date.

- Select the Attribute Column
- Go to the **Add Column** tab
- Click **Date → Year → Year**
- This will create a new column with the year value

Formula: = Table.AddColumn(#"#5 - Changing Column Type to Date", "Year", each Date.Year([Attribute]), Int64.Type)

#7 - Changed Data Type to Number – This step converts the **Value** Column into a numeric data type.

- Select the **Value** column
- Go to the **Transform** tab
- Click **Data Type**
- Choose **Decimal Number**

Back to Table Of Contents

Formula: = Table.TransformColumnTypes(#"#6 - Inserting Year Column",{{"Value", type number}})

 # 8 - Grouping Rows By Year – This step groups the data by **Year** and **Zip Code** and then calculates the home value.

- Go to the **Transform** tab
- Click **Group By**
- Select **Advanced**
- Add grouping columns:
  - **Year**
  - **Zip Code**
- Under New column:
  - Name: **Avg Home Value**
  - Operation: **Average**
  - Column: **Value**
  - Click OK

Formula: = Table.Group(#"#7 - Changed Data Type to Number", {"Year", "Zip Code"}, {{"Median Home Value ", each List.Average([Value]), type text}})

 # 9 - Changed Two Columns Data Type to Numeric– This step converts the **Zip Code** and **Median Home Value** columns into a whole number data type.

- Select the **Zip Code** column
- Hold **Ctrl** and select the **Median Home Value** column
- Go to the **Transform** tab
- Click **Data Type**
- Choose **Whole Number**

Formula: = Table.TransformColumnTypes(#"# 8 - Grouping Rows By Year ",{{"Zip Code", Int64.Type}, {"Median Home Value", Int64.Type}})

 # 10 - Filter Years to 2015-2025 – This step filters the dataset to only include records from the year 2015 - 2025

- **Select the Year column**
- Click the filter dropdown

Back to Table Of Contents

- Choose **Number Filters → Between**
- Enter:
- **2015** - is greater than or equal to
- **2025** - is less than or equal to
- Make sure condition is set to **AND**
- Click **OK**

Formula: = Table.SelectRows(#"# 9 - Changed Two Columns Data Type to Numeric", each [Year] >= 2015 and [Year] <= 2025)

Linear Regression Model – Measures in Power BI

Median Home Value Linear Regression

This measures calculates a linear regression model to estimate and predict median home values based on the relationship between year and home prices. Based on an article from https://www.sqlbi.com/articles/implementing-linear-regression-in-power-bi/

How it was created

- Go to the **Modeling** tab
- Click **New Measure**
- Enter the DAX formula below for the regression model
- Use **Median Home Value** as the Y value
- Use **Year** as the X value
- Use the slope and intercept to calculate the predicted home value

Formula:

```
MedianHomeValueLinearRegression =
// This measure uses LINESTX to create a linear regression model.
// Guidance was based on SQLBI's article on implementing linear regression in Power BI.
VAR line =
   LINESTX (
      ALL ( 'Median_Home_Values' ), // Uses all rows from Median_Home_Values to calculate the regression line
      'Median_Home_Values'[Median Home Value], // Y value: value being predicted
      'Median_Home_Values'[Year] // X value: independent variable
   )
// Gets the slope, which shows the yearly rate of change
VAR slope =
   MAXX ( line, [Slope1] )
// Gets the intercept, which is the starting point of the regression line
```

Back to Table Of Contents

```
VAR intercept =
    MAXX ( line, [Intercept] )
// Gets the current year from the Year_Table
VAR x =
    MAX ( Year_Table[Year] )
// Returns the predicted median home value using y = mx + b
RETURN
    intercept + slope * x
```

Breakdown Of Formula

1. **Creating the regression line**

```
VAR line =
    LINESTX (
        ALL ( 'Median_Home_Values' ), // Using all the data from Median_Home_Values in
order to be used for regression
        'Median_Home_Values'[Median Home Value], // Y Value
        'Median_Home_Values'[Year] // X Value
    )
```

This step creates a regression line using the **LINESTX** function. It uses all the rows from the **Median_Home_Values** table. The **Median Home Value** column is the value that is being predicted and the **Year** column is the value that is being used to predict it.

2. **Get the slope**

```
VAR slope =
    MAXX ( line, [Slope1] )
```

This step gets the slope from the regression model. The slope will show how much the value in median home changes each year.

3. **Get the intercept**

```
VAR intercept =
    MAXX ( line, [Intercept] )
```

This step gets the intercept from the regression model. The intercept will be used to be the starting point of the linear regression equation.

Back to Table Of Contents

4. Get the current year

```
VAR x =
    MAX(Year_Table[Year])
```

This step gets the current year from the Year_Table. This allows the measure to calculate the predicted value for each year in the visual.

5. Return the predicted value

```
RETURN
    intercept + slope * x
```

This measure uses the relationship between **Year** and **Median Home Value** to estimate the predicted home value for each of the year on the visual.

MedianHouseholdIncomeLinearRegression

These measures calculated a linear regression model to estimate and predict median household income based on the relationship between year and income. Based on an article from https://www.sqlbi.com/articles/implementing-linear-regression-in-power-bi/

How it was created

- Go to the **Modeling** tab
- Click **New Measure**
- Enter the DAX formula below for the regression model
- Use **Median Household Income** as the Y value
- Use **Year** as the X value
- Use the slope and intercept to calculate the predicted home value

Formula:

```
MedianHouseholdIncomeLinearRegression =
// used https://www.sqlbi.com/articles/implementing-linear-regression-in-power-bi article to
learn and use the LINESTX DAX function for linear regression model
VAR line =
    LINESTX (
        ALL ( 'Median_Household_Income' ), // Uses all income data for regression
        'Median_Household_Income'[Median Household Income], // Y Value
        'Median_Household_Income'[Year] // X Value
    )
// Gets the slope
VAR slope =
    MAXX ( line, [Slope1] )
// Gets the intercept
VAR intercept =
```

Back to Table Of Contents

```
    MAXX ( line, [Intercept] )
// Current X-value
VAR x =
    MAX(Year_Table[Year])
// Returns the linear regression
RETURN
    intercept + slope * x
```

1. **Creating the regression line**

```
VAR line =
    LINESTX (
        ALL ( 'Median_Household_Income' ), // Uses all income data for regression
        'Median_Household_Income'[Median Household Income], // Y Value
        'Median_Household_Income'[Year] // X Value
    )
```

This step creates a regression line using the **LINESTX** function. It uses all the rows from the **Median_Household_Income** table. The **Median_Household_Income** column is the value that is being predicted and the year columns is the value that is being used to predict it.

2. **Get the slope**

```
VAR slope =
    MAXX ( line, [Slope1] )
```

This step gets the slope from the regression model. The slope will show how much the value in median household income changes over year.

3. **Get the intercept**

```
VAR intercept =
    MAXX ( line, [Intercept] )
```

This step gets the intercept form the regression model. The intercept will be used to be the starting point of the linear regression equation.

4. **Get the current year**

```
VAR x =
    MAX(Year_Table[Year])
```

This step gets the current year from the Year_Table. This allows the measure to calculate the predicted income value for each year in the visual.

Back to Table Of Contents

5. **Return the predicted value**

```
RETURN
    intercept + slope * x
```

This measure uses the relationship between **Year** and **Median Household Income** to estimate the predicted median household Income value for each year on the visual.

Measures

This section will explain the main custom measures and calculated fields that were created in Power BI. These measures were used in order to analyze housing affordability , compare the growth between household income and home values and help support the visuals on the dashboard.

Affordability Ratio

This measure calculates how many times higher the price of a median home value is compared to the median household income. The measure to see how many times a home is more than the salary.

Why was it used?

This measure was used to show how affordable homes are compared to income. The higher the ratio, the less affordable the home is.

- Go to modeling tab
- Click **New Measure**
- Divide the median home value by the median household income.

```
Affordability Ratio =
// Calculates the affordability ratio by dividing the home value by household income
// Shows how many times higher the home value is compared to household income
DIVIDE(
    MAX('Median_Home_Values'[Median Home Value]),
    MAX(Median_Household_Income[Median Household Income])
)
```

Affordability Ratio – Regression Values

This measure calculates how many times higher the price of a predicted median home value is compared to the predicted median household income. The measure to see how many times a home is more than the salary.

Back to Table Of Contents

Why was it used?

This measure was used to show how affordable homes are compared to income. The higher the ratio, the less affordable the home is.

- Go to **Modeling** tab
- Click **New Measure**
- Divide the predicted median home value by the predicted median household income.

```
Affordability Ratio - Regression Values =
// Calculates the affordability ratio by dividing the predicted home value by the predicted
household income
DIVIDE(
   [Median Home Value Linear Regression],
   [Median Household Income LinearRegression]
)
```

Home Growth % Since 2015

This measure calculates how much median home values have increased since 2015.

Why was it used?

This measure was used as a **KPI** on the **Linear Regression Model Analysis – Detailed** page of the dashboard to show the percentages growth of Home Values since 2015 then compared to the linear regression predicted value.

- Go to the **Modeling** tab
- Click **New Measure**
- Create a variable to store the 2015 median home value
- Subtract the 2015 value from the selected year's predicted home value
- Divide that difference by the 2015 median home value

```
Home Growth % Since 2015 =

// Create a variable called Home 2015 to get the actual median home value from 2015
VAR Home2015 =
   CALCULATE(
      AVERAGE('Median_Home_Values'[Median Home Value]),
      ALL(Year_Table),
      'Median_Home_Values'[Year] = 2015
   )

RETURN
// Calculates growth from the 2015 actual home value to the selected year's predicted home
value
```

Back to Table Of Contents

```
    DIVIDE(
        [Median Home Value Linear Regression] - Home2015,
        Home2015)
```

Home Growth % Overview

This measure calculates the total percent growth in median home values from 2015 – 2025.

Why was it used?

This measure was used as a **KPI** on the **Overview** page of the dashboard to show how much home values have increased from 2015 – 2025. This helps summarize the overall change from the start and end year dates. Go to the **Modeling** tab

- Go to the **Modeling** tab
- Click **New Measure**
- Create a variable for the 2015 median home value
- Create a variable for the 2025 median home value
- Subtract the 2015 value from the 2025 value
- Divide the difference by the 2015 value

```
Home Growth % Overview =
// Creating a variable called Home 2015 to get Median Home Value from 2015
VAR Home2015 =
    CALCULATE(
        AVERAGE('Median_Home_Values'[Median Home Value]),
        'Median_Home_Values'[Year] = 2015
    )
// Creating a variable called Home2025 to get Median Home Value from 2025
VAR Home2025 =
    CALCULATE(
        AVERAGE('Median_Home_Values'[Median Home Value]),
        'Median_Home_Values'[Year] = 2025
    )
// Calculates the median home value in 2025 by the home price from 2015

RETURN
    DIVIDE(
        Home2025 - Home2015,
        Home2015
    )
```

Income Growth % Since 2015

This measure calculates how much the median Household Income have increased since 2015.

Back to Table Of Contents

Why was it used?

This measure was used as a **KPI** on the **Linear Regression Model Analysis – Detailed** page of the dashboard to show the percentages growth compared to the linear regression predicted value.

- Go to the **Modeling** tab
- Click **New Measure**
- Create a variable to store the 2015 median household income value
- Subtract the 2015 value from the selected year's predicted income value
- Divide that difference by the 2015 income value

```
Income Growth % Since 2015 =
// Saves Household income from 2015 into a Variable
VAR Income2015 =
   CALCULATE(
      AVERAGE(Median_Household_Income[Median Household Income]),
      ALL(Year_Table),
      Median_Household_Income[Year] = 2015
   )

RETURN
// Calculates the percent growth from 2015 income to the selected year's predicted income
   DIVIDE(
      [Median Household Income LinearRegression] - Income2015,
      Income2015
   )
```

Down Payment (5%)

This measure estimates the amount needed for a 5% down payment based on the median home price.

Why was it used?

This was used to show how much money a buyer would need for first-time homebuyer rate of 5% downpayment of a house.

- Go to **Modeling** tab
- Click **New Measure**
- Multiply the **Median Home Value** by 0.05

```
Down Payment (5%) =
// Calculates 5% of the cost for the Median Home Value
'Median_Home_Values'[Median Home Value] * 0.05
```

Back to Table Of Contents

Down Payment (20%)

This measure estimates the amount needed for a 20% down payment based on the median home price.

Why was it used?

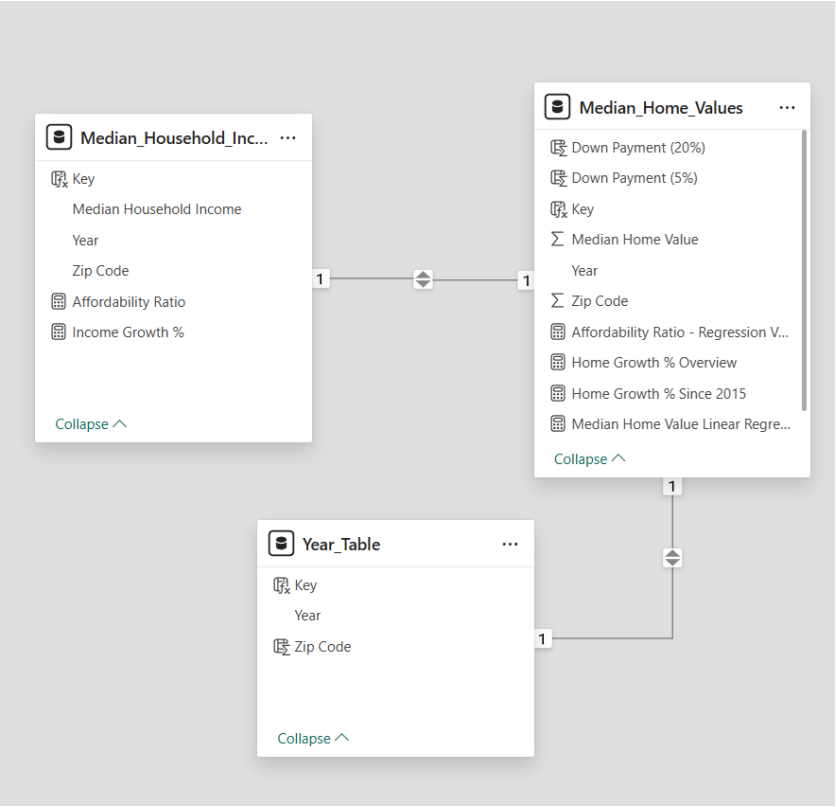
This was used to show how much money a buyer would need for the traditional 20% downpayment of a house.

- Go to modeling tab
- Click New Measure
- Multiply the Median Home Value by 0.20

```
Down Payment (20%) =
// Calculates 20% of the cost for the Median Home Value
'Median_Home_Values'[Median Home Value] * 0.20
```

Back to Table Of Contents

Data Model and Table Relationships



This section explains how the main tables were connected in Power BI. Relationships that were created in order for the dashboard to compare both Median Home Value and Median Household Income by the same year and Zip Code.

Key Column

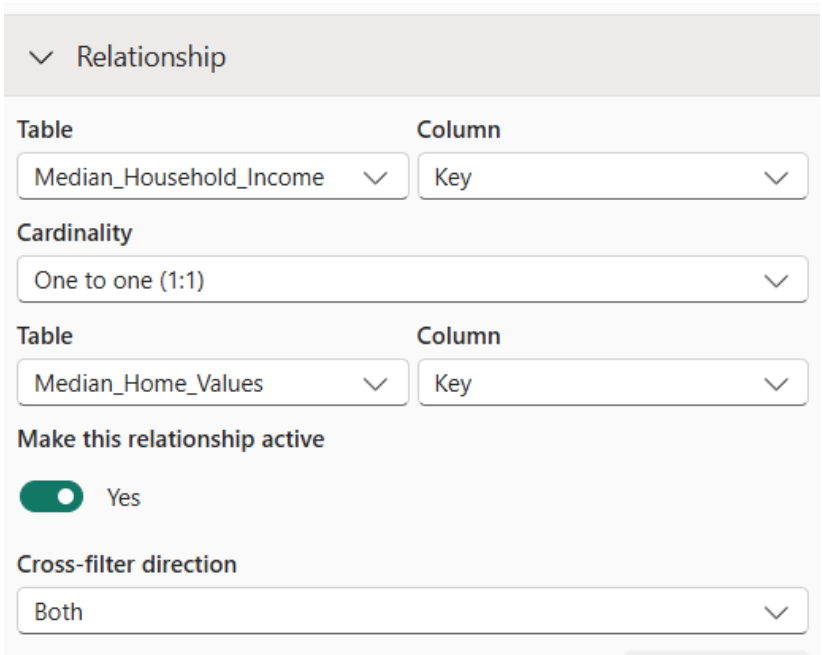
This **Key** column was created in order to combine the **Year** and **Zip Code** fields into one unique identifier. This helps match up the records between both tables that share the same year and zip code.

Formula: Key = [Year] & "-" & [Zip Code]

Back to Table Of Contents

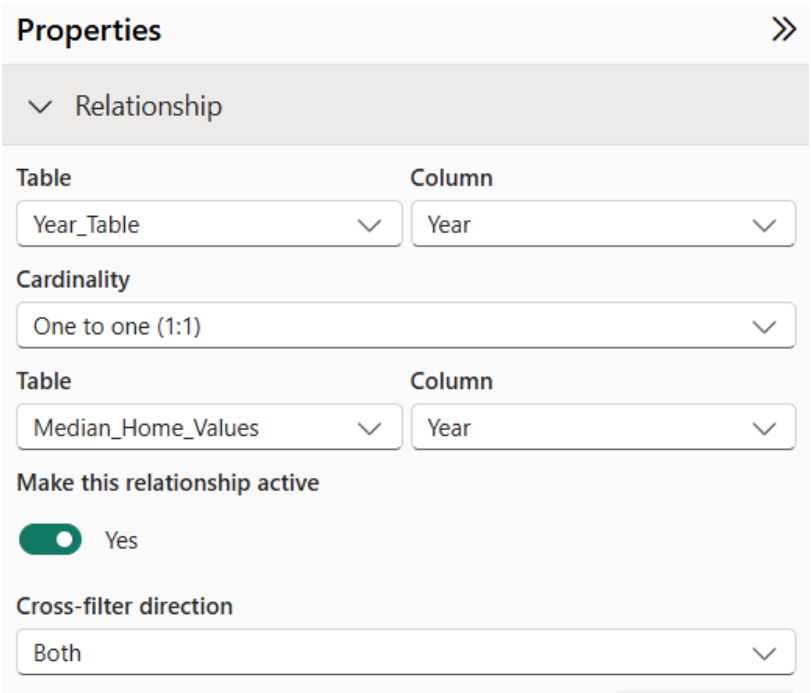
Relationship Between Household Income and Median Home Values

The two tables Median_Household_Income and the Median_Home_Values were connected with the **Key** column. This allows home value and income data to be able to be compared together on the dashboard visuals.



Relationship Between Year Table and Median Home Values

The **Year_Table** connected to the **Median_Home_Values** table using the **Year** column. This was used in order to support year-based filtering with the regression models.



Back to Table Of Contents